6.2: The Kernel and Range of a Linear Transformation

MATH 2360

Definitions

1. **Kernel of a Linear Transformation:** Let $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ be a linear transformation. Then the set of all vectors $\vec{v}$ in $\mathcal{V}$ that satisfy $\mathcal{T}(\vec{v}) = \vec{0}$ is the **kernel** of $\mathcal{T}$, denoted $\ker(\mathcal{T})$. (Also called the **nullspace** of $\mathcal{T}$.)

Examples: Find the kernel of each of the following:

1) If $\mathcal{T}: M_{3,2} \to M_{2,3}$ such that $\mathcal{T}(A) = A^T$.

2) If $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ is the zero transformation.

3) If $\mathcal{T}: \mathcal{V} \to \mathcal{V}$ is the identity transformation.

4) If $\mathcal{T}: \mathbb{R}^3 \to \mathbb{R}^3$ such that $\mathcal{T}(x, y, z) = (x, y, 0)$.

5) Find the kernel of $\mathcal{T}: \mathbb{R}^2 \to \mathbb{R}^2$ represented by $\mathcal{T}(x, y) = (x + 2y, y - x)$.

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2. **Thm:** The kernel of a linear transformation $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ is a subspace of $\mathcal{V}$.

3. **Corollary:** Let $\mathcal{T}: \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation $\mathcal{T}(\vec{x}) = A\vec{x}$. Then the kernel of $\mathcal{T}$ is equal to the solution space of $A\vec{x} = \vec{0}$.

Example:

6) Find a basis for $\ker(\mathcal{T})$ if $\mathcal{T}(\vec{x}) = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 2 \end{bmatrix}\vec{x}$.

4. **Thm:** The range of a linear transformation $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ is a subspace of $\mathcal{W}$.
   Recall: range$(\mathcal{T}) = \{\mathcal{T}(\vec{v}): \forall \vec{v} \in V\}$

5. **Corollary:** Let $\mathcal{T}: \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation $\mathcal{T}(\vec{x}) = A\vec{x}$. Then the range of $\mathcal{T}$ is equal to the column space of A.

Example:

7) Find a basis for range$(\mathcal{T})$ for the transformation in the previous problem.
   $\mathcal{T}(\vec{x}) = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 2 \end{bmatrix}\vec{x}$.

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6. **Rank and Nullity:** Let $\mathcal{T}: \mathcal{V} \rightarrow \mathcal{W}$ be a linear transformation. The dimension of the kernel of $\mathcal{T}$ is the **nullity** of $\mathcal{T}$ (denoted $nullity(\mathcal{T})$). The dimension of the range of $\mathcal{T}$ is the **rank** of $\mathcal{T}$ (denoted $rank(\mathcal{T})$). If $\mathcal{T}: \mathcal{V} \rightarrow \mathcal{W}$ and $\mathcal{V}$ is n-dimensional, then

$$rank(\mathcal{T}) + nullity(\mathcal{T}) = n \qquad (dim(range) + dim(kernel) = dim(domain))$$

Example:

8) Let $\mathcal{T}: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation.

a) Find $rank(\mathcal{T})$ when $nullity(\mathcal{T}) = 3$.

b) Find the dimension of the kernel of $\mathcal{T}$ when the dimension of the range is 2.

c) Find $rank(\mathcal{T})$ when $\ker(\mathcal{T}) = \{\vec{0}\}$

7. **One-to-One Thm:** Let $\mathcal{T}: \mathcal{V} \rightarrow \mathcal{W}$ be a linear transformation. Then $\mathcal{T}$ is **one-to-one** iff $\ker(\mathcal{T}) = \{\vec{0}\}$.

8. **Onto Thm:** Let $\mathcal{T}: \mathcal{V} \rightarrow \mathcal{W}$ be a linear transformation where $\mathcal{W}$ is finite dimensional. Then $\mathcal{T}$ is **onto** iff the rank of $\mathcal{T}$ is equal to the dimension of $\mathcal{W}$. (Every element in $\mathcal{W}$ has a preimage in $\mathcal{V}$.)

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9. **Thm:** Let $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ be a linear transformation where $\mathcal{V}$ and $\mathcal{W}$ are both dimension n. Then $\mathcal{T}$ is one-to-one iff it is onto.

10. **Isomorphism Thm:** A linear transformation $\mathcal{T}: \mathcal{V} \to \mathcal{W}$ that is one-to-one and onto is an isomorphism. ($\mathcal{V}, \mathcal{W}$ are isomorphic)

11. **Thm:** Two finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ are isomorphic iff they are of the same dimension.

Example:

9) Determine whether the following vector spaces are isomorphic to each other:

$$\mathbb{R}^4, M_{4,1}, M_{2,2}, P_3, \{(x_1, x_2, x_3, x_4, 0)\}$$

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